

Chapter 6, Problem 8PP

Problem

If the current through a 1-mH inductor is $i(t) = 90 \sin(200t)$ mA, find the terminal voltage and the energy stored.

Step-by-step solution

Step 1 of 2

The expression for the inductor voltage is,

$$v(t) = L \frac{di(t)}{dt}$$

Substitute 1 mH for L and $90 \sin(200t)$ mA for $i(t)$.

$$\begin{aligned} v(t) &= (1 \times 10^{-3}) \frac{d}{dt} [90 \sin(200t) \times 10^{-3}] \\ &= (1 \times 10^{-3}) 90 \times 10^{-3} \times 200 \cos(200t) \\ &= 18 \times 10^{-3} \cos(200t) \text{ V} \\ &= 18 \cos(200t) \text{ mV} \end{aligned}$$

Thus, the inductor voltage is $18 \cos(200t) \text{ mV}$

Step 2 of 2

The Energy stored in the inductor is,

$$w = \frac{1}{2} Li^2$$

Substitute 1 mH for L and $90 \sin(200t)$ mA for $i(t)$.

$$\begin{aligned} w &= \frac{1}{2} (1 \times 10^{-3}) (90 \sin(200t) \times 10^{-3})^2 \\ &= \frac{1}{2} (1 \times 10^{-3}) \times 8100 \times 10^{-6} \sin^2(200t) \\ &= 4.05 \sin^2(200t) \mu\text{J} \end{aligned}$$

Thus, the energy stored in the inductor is $4.05 \sin^2(200t) \mu\text{J}$